

Diabetes Risk Prediction Project Documentation

Introduction and Objective

DM (Diabetes Mellitus) is a major global metabolic health challenge that requires early screening and proactive clinical intervention. The objective of this project was to create a model that accurately predicts diabetes risk using raw data consisting of medical diabetic predisposing factors records. By leveraging patient vitals such as Plasma Glucose Concentration, BMI, Insulin level, and Age, the project demonstrates how clinical data science can assist healthcare providers in diagnostic triage.

Data Cleaning

Empty cells in the columns specifically **Glucose, Blood Pressure, Skin Thickness, Insulin, and BMI** were identified as missing data, as no measurements in these categories are biologically incompatible with life.

Missing values were handled by using **median imputation** directly on the affected columns. The median was selected over the mean because it is mostly not affected by extreme outliers, ensuring a natural distribution of data.

Exploratory Data Analysis (EDA)

A comprehensive visual exploration was performed on the training set to map out patient distribution and isolate diagnostic indicators.

Insight 1: Distribution Count Plot

The count plot revealed that roughly **65%** of the patients were healthy (0), while **35%** had a positive diabetes diagnosis (1).

Insight 2: Glucose Box Plot by Diagnostic Outcome

This box plot split by diagnostic outcome showed an undeniable diagnostic gap. It showed that healthy individuals maintained a tightly bound, stable glucose concentration with a median of approximately **110 mg/dL**, while the diabetic individuals had an elevated and a broader distribution with a median around **140 mg/dL**, confirming blood sugar as a major model indicator.

Insight 3: Scatter Plot of BMI and Glucose Concentration by Diabetes Outcome

Plotting the glucose concentration directly with the BMI using a color hue revealed a clear clinical danger zone. While healthy patients are densely clustered in the lower

left quadrant (**Glucose < 120 mg/dL, BMI < 35 kg/m²**), the upper right quadrant—where high glucose and obesity intersect—was heavily dominated by positive diabetic outcomes.

Insight 4: Age Distribution and Diabetes Risk with Stacked Histogram

The age distribution chart showed that while the study population was highly skewed toward younger adults (**20–30**), the proportion of diabetic outcomes increases drastically with age. The peak risk density shifted heavily within the **30–50** age bracket.

Insight 5: Features Correlation Heatmap

The heatmap shows a relationship across all metrics, with **glucose** yielding the highest positive correlation (**0.48**), followed by **BMI** and **age**. This shows they have a high effect on diabetic outcomes.

Data Splitting

To avoid data leakage, the cleaned dataset was split into an **80% training set (614 patients)** and a **20% testing set (154 patients)**.

Data Scaling

Because logistic regression relies on mathematical optimization, features with large numerical ranges could overpower features with smaller ranges. To eliminate this dimensional bias, **Standard Scaling (StandardScaler)** was applied to the 8-dimensional feature matrix. This step shifted all metrics to a shared scale with a mean of 0 and a standard deviation of 1, placing indicators like Glucose and BMI on an equal playing field.

Hyperparameter Optimization (Grid Search)

To maximize the clinical utility of the model, a **Grid Search Cross-Validation (GridSearchCV)** pipeline was deployed using 5-fold cross-validation. Rather than settling for factory-default settings, the model systematically evaluated different values for the regularization strength parameter ('C': [0.01, 0.1, 1, 10, 100]) and optimization solvers (liblinear and lbfgs). The grid search was explicitly configured to optimize for **Recall** to prioritize patient safety by minimizing missed diagnoses.

Model Training and Feature Importance

The optimized Logistic Regression model was trained on the standardized training set. Following optimization, the mathematical weights (coefficients) were extracted to audit the model's underlying clinical logic. The feature importance analysis revealed that **Glucose**, **BMI**, and **Age** yielded the highest positive coefficients, aligning perfectly with established epidemiological risk factors for Type 2 diabetes.

Model Evaluation and Confusion Matrix

The finalized model was subjected to its final exam using the untouched test set of 154 patients (100 confirmed healthy and 54 diabetic). The model achieved a strong overall diagnostic accuracy of **79%** (correctly classifying approximately 4 out of every 5 patients).

To evaluate the model's safety for preliminary clinical screening, the final predictions were mapped into a confusion matrix and classification report:

- **True Negative (TN):** 85 (Healthy patients correctly identified)
- **False Positive (FP):** 15 (Healthy patients flagged as at-risk)
- **False Negative (FN):** 17 (Diabetic patients missed by the model)
- **True Positive (TP):** 37 (Diabetic patients correctly identified)

Performance Metrics Interpretation

- **Precision (71%):** Out of all individuals flagged as "High Risk" by the algorithm, 71% were true positive cases. This reliable rate helps manage the clinical pipeline without creating excessive false alarms for healthy individuals.
- **Recall (69%):** The model successfully captures 69% of the positive diabetic outcomes. Increasing this sensitivity remains a primary focus for future optimization, as an undiagnosed patient faces severe long-term systemic complications.
- **F1-Score (0.70):** Represents a stable and balanced harmonic mean between precision and sensitivity, proving the model is resilient against the dataset's mild class imbalance.

Conclusion

This project successfully completed a full end-to-end healthcare predictive pipeline. By utilizing global median cleaning, standard scaling, and hyperparameter optimization via cross-validation, a robust Logistic Regression framework was trained, achieving a final test accuracy of **79%**.

Recommendations

The trained Logistic Regression model achieves an overall accuracy of **79%** and a precision of 71%, making it highly viable for deployment as an automated clinical triage assistant within healthcare facilities.

It is recommended that this model be integrated into the clinic's digital reception workflow or an interactive dashboard to automatically flag high-risk individuals the moment their basic vitals are logged. Because the majority of patients flagged by the algorithm will be true positive cases, the hospital can confidently utilize the model to fast-track at-risk individuals directly to confirmatory laboratory testing, optimizing resource allocation and reducing patient waiting times.

However, because the baseline model exhibits a final recall of 69%, it will let approximately 31% of true diabetic instances slip through undetected as false negatives. For this reason, the model must never be used as a standalone diagnostic tool or to give a final medical clearance; instead, it must operate strictly as a secondary decision-support system where human clinical judgment and patient symptoms always maintain the final authority.

To improve clinical safety in future iterations, the classification threshold should be adjusted to maximize recall, and more specific diagnostic features, such as HbA1c percentages and family medical history, should be integrated into the dataset.